REPORT

Title

Prolog Program for Fruit and Its Color Using Backtracking

Aim

To implement a Prolog program to represent **fruits and their colors** and demonstrate the concept of **backtracking**.

Problem Statement

Create a Prolog database containing different fruits and their colors. The program should retrieve the color of a particular fruit or display all fruit-color combinations using Prolog's backtracking mechanism.

Requirements

- SWI-Prolog
- Computer system
- Prolog source file (`exp10.pl`)

Procedure

1. Open **SWI-Prolog**.
2. Create a new file named `exp10.pl`.
3. Enter the fruit and color facts.
4. Save the program.
5. Load the program using `[exp10].`.
6. Query the database using `fruit(Fruit, Color).`.
7. Press `;` to activate backtracking and display the next solution.
8. Continue pressing `;` until all solutions are displayed.

Program

```
fruit(apple, red).
fruit(banana, yellow).
fruit(orange, orange).
fruit(grapes, green).
fruit(mango, yellow).
fruit(watermelon, green).
fruit(strawberry, red).
```

Sample Output

Query:

```
?- [exp10].
true.

?- fruit(Fruit, Color).
```

Output:

```
Fruit = apple,
Color = red ;

Fruit = banana,
Color = yellow ;

Fruit = orange,
Color = orange ;

Fruit = grapes,
Color = green ;

Fruit = mango,
Color = yellow ;

Fruit = watermelon,
Color = green ;

Fruit = strawberry,
Color = red.
```

For a particular fruit:

```
?- fruit(mango, Color).
```

Output:

```
Color = yellow.
```

Conclusion

Thus, the **fruit and color database was successfully implemented using Prolog backtracking**. Prolog automatically searches for alternative facts when `;` is pressed, allowing all matching fruit-color combinations to be retrieved.